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ABSTRACT

Packaged food products and personal care items frequently contain complex ingredient lists composed of chemical names, CI codes, E-numbers, and INS numbers that are difficult for ordinary consumers to understand. Although regulatory bodies mandate ingredient disclosure, the technical terminology used often prevents users from interpreting the information meaningfully. As a result, many consumers ignore labels entirely or make uninformed purchasing decisions that may expose them to allergens, artificial additives, or substances associated with potential health concerns. The “AI-Powered Ingredient Scanner & Risk Analyzer” is designed to address this challenge by offering an intelligent, user-friendly system capable of scanning and decoding ingredient labels. The application utilizes Optical Character Recognition (OCR) to extract printed text from uploaded product images and applies rule-based analysis to identify chemical codes and additives. The extracted ingredients are translated into plain-English descriptions, accompanied by safety classifications and contextual warnings. The platform further enhances usability by supporting user-selectable safety profiles such as children, pregnant women, elderly individuals, and allergen-sensitive users. Visual dashboards, traffic light indicators, and product grading mechanisms simplify risk communication. By integrating OCR, Python-based processing, SQLite databases, and visualization tools, the system promotes consumer awareness, encourages safer choices, and contributes to preventive health consciousness.

1. INTRODUCTION

➤ Literature Survey

Ingredient Labels and Consumer Awareness

Ingredient labeling has become a fundamental aspect of modern consumer protection, particularly in packaged foods, cosmetics, and healthcare products. Regulatory authorities such as FSSAI require manufacturers to disclose product contents to ensure transparency and informed decision-making. However, research in consumer studies reveals that despite the availability of ingredient lists, comprehension levels among users remain low. The technical language used, including chemical compounds, abbreviations, and additive codes, often discourages consumers from engaging with label information.

Many additives are represented using standardized identifiers such as E-numbers (E211), CI codes (CI 16035), or INS numbers (INS 110), which lack intuitive meaning for non-experts. Studies indicate that consumers are more likely to respond to simplified and human-readable explanations rather than scientific nomenclature. Additionally, literature in public health communication highlights that effective risk messaging requires clarity, personalization, and visual reinforcement.

With advancements in artificial intelligence, OCR and NLP technologies have enabled automated extraction and interpretation of textual information from images. Smart label scanning systems

demonstrate the potential to enhance consumer literacy by converting technical content into understandable insights. Research suggests that combining AI with visual analytics improves trust, usability, and behavioral adoption, particularly in health-related applications.

➤ Limitations of Current Systems

Despite the emergence of food-scanning and nutritional analysis applications, several limitations persist in current systems. A majority of existing solutions rely primarily on barcode scanning, which is dependent on centralized product databases. These databases may lack entries for regional brands, newly launched products, or items without barcodes. Consequently, barcode-only systems fail to provide universal usability. Furthermore, many applications emphasize macronutrient breakdowns (calories, fats, sugars) while offering limited interpretation of ingredient-level safety concerns. Additives, preservatives, artificial colors, and allergens are often not explained in sufficient depth. Research in consumer safety indicates that ingredient-specific risk awareness is critical for vulnerable groups such as children, pregnant women, and allergy-sensitive individuals. Affordability also presents a challenge. Advanced features such as ingredient decoding, detailed warnings, and personalized recommendations are frequently restricted behind premium subscriptions. Privacy concerns are equally significant. Studies on digital ethics stress that users are reluctant to adopt scanning tools if data collection practices lack transparency. Additionally, many platforms lack offline functionality, making them unreliable in low-connectivity environments. These gaps highlight the need for a privacy-conscious, OCR-enabled system capable of delivering deeper ingredient intelligence beyond conventional nutritional summaries.

➤ Proposed Work

Core Design of the Ingredient Scanner & Risk Analyzer

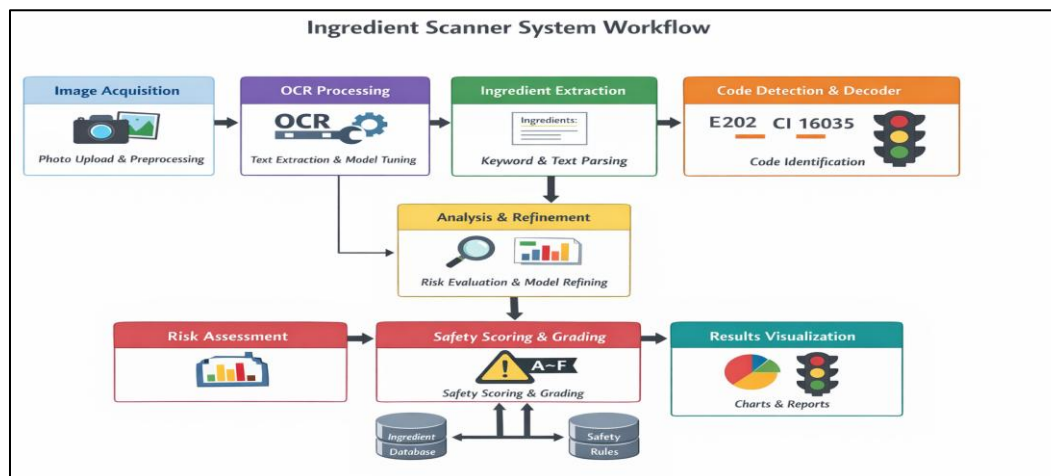


Fig1. System diagram

To overcome the limitations of existing platforms, this project proposes the development of an AI-Powered Ingredient Scanner & Risk Analyzer using OCR and Python-based processing. The system enables users to upload or capture images of product labels, which are processed through OCR engines such as EasyOCR or PaddleOCR. These technologies extract printed textual information even from varied fonts and packaging layouts. Following OCR extraction, the text undergoes preprocessing steps including noise filtering, normalization, and segmentation. Pattern recognition techniques using regular expressions identify ingredient sections, chemical codes, and additive identifiers. The extracted ingredients are then cross-referenced with a locally stored SQLite database containing additive descriptions, risk classifications, regulatory status, and safety warnings. A rule-based risk evaluation engine assigns safety scores based on ingredient categories and profile-specific sensitivity. The system produces cumulative risk ratings and generates an overall product grade (A–F). By translating chemical identifiers into plain-English explanations, the platform significantly improves consumer understanding.

Unlike conventional barcode-based systems, this solution emphasizes universal label scanning, chemical code decoding, offline capability, and visual risk communication.

Extended Features and Vision

Beyond ingredient extraction and decoding, the proposed platform aspires to enhance consumer engagement and decision-making through intelligent design. The inclusion of user-selectable safety profiles ensures contextual analysis. For example, ingredients considered safe for adults may trigger warnings under child or pregnancy modes. This personalization aligns with research emphasizing targeted health communication. The visualization dashboard presents safety insights using traffic light indicators, pie charts, and grading systems. Such visual cues reduce cognitive load and improve interpretability. The scan history module enables users to review previously analyzed products, supporting comparison and behavioral reflection. The system prioritizes accessibility and privacy by eliminating mandatory user accounts and storing scan data locally. Offline-first functionality ensures reliability without continuous internet dependence. Report export capabilities allow users to generate PDF summaries for documentation or sharing. Collectively, these features transform the application into a practical consumer safety assistant rather than a simple scanning tool.

2. METHODOLOGY

The project follows a modular and systematic design approach to ensure clarity, maintainability, and scalability. Each functional component is developed independently while maintaining seamless integration within the Streamlit-based interface. The workflow begins with image acquisition, followed by OCR processing, text extraction, ingredient identification, decoding, risk analysis, and visualization.

The system consists of the following modules:

- Image Acquisition Module:
 - Upload product label images through Streamlit interface
 - Support scanning of stored/gallery images
 - Image enhancement and preprocessing using PIL
 - Standardization of image formats for OCR compatibility
- OCR Processing Module:
 - Text extraction using EasyOCR / PaddleOCR
 - Multi-language printed text recognition
 - Noise removal and correction of OCR errors
 - Output generation as structured raw text
- Ingredient Extraction Module:
 - Detection of “Ingredients” keyword
 - Extraction of relevant lines following header
 - Cleaning, tokenization, normalization
 - Removal of irrelevant symbols/noise
- Code Detection & Decoder Module:
 - Regex detection of CI codes, E-numbers, INS numbers
 - Batch/expiry code identification
 - Plain-English translation of chemical identifiers
 - Mapping with database records
- Risk Analysis Module:
 - Ingredient lookup using SQLite
 - Risk classification (Low / Medium / High)
 - Profile-specific filtering logic
 - Cumulative risk scoring and grading
- Visualization Module:
 - Traffic light safety indicators
 - Plotly pie charts and summaries
 - Risk percentage display
 - Final product safety grade

- **Software Requirements:**
 - Frontend: Streamlit
 - Backend: Python
 - OCR: EasyOCR / PaddleOCR
 - Database: SQLite
 - Visualization: Plotly
 - Image Processing: PIL
- **Hardware Requirements**
 - Processor: Intel i3 / AMD equivalent or higher
 - RAM: 4 GB minimum (8 GB recommended)
 - Storage: 256 GB HDD/SSD
 - Internet: Optional (offline supported)
 - Camera: Optional

3. OUTCOMES

- **Student-Specific Outcomes**
 - Hands-on experience with OCR and AI tools
 - Practical knowledge of Python & Streamlit
 - Understanding of text preprocessing & regex
 - Experience in database design (SQLite)
 - Exposure to visualization libraries (Plotly)
 - Enhanced UI/UX design skills
- **Project-Specific Outcomes**
 - Functional OCR-based ingredient scanner
 - Accurate chemical code detection & decoding
 - Ingredient-level safety interpretation
 - Risk scoring and product grading system
 - Interactive visualization dashboard
 - Improved consumer awareness and safety tool

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